

# Supplement 2: Definitions, Theorems and Worked Examples

This supplement holds the formal apparatus of the main paper: the definitions that fix the object, the two theorems and their corollaries that classify it, and the worked examples that carry a reader through one member of the family by hand. The main paper states each result in prose where it is used and points here for the statement, the proof, and the arithmetic. Numbering matches the order of first use in the paper.

## Definition 1: Writings, elementary writes, and propagators

A *writing* is any function  $w: \{0, \dots, 7\} \rightarrow \{0, \dots, 7\}$  on the truth-table positions. Its elementary update at source position  $p$  is

$$(U_{w,p}g)(q) = \begin{cases} \neg g(p), & q = w(p), \\ g(q), & q \neq w(p). \end{cases}$$

Thus one update complements source bit  $p$  and writes it to target  $w(p)$ . If  $w$  is a bijection—a permutation  $\sigma \in S_8$ —all eight writes can be performed simultaneously without collisions, giving the *permutation-propagator*  $P_\sigma$  in Equation (1). Its fixed bytes are exactly the bytes fixed by every  $U_{\sigma,p}$ . For a non-injective writing, individual updates  $U_{w,p}$  remain well defined, but a simultaneous  $P_w$  requires a collision-resolution convention; such a writing also has *gaps*, or positions targeted by no source.

## Observation 1: Permutations form a small bijective core

Permutations are only the bijective core of a far larger class: there are  $8! = 40,320$  permutations but  $8^8 = 16,777,216$  writings, so the permutations are 0.24% of the whole. The cycle-structure algebra of *The Classification* is available only on this core. The historical operator  $[00123456]$  (Figure 3) is itself a non-bijective writing—bits 0 and

1 both write to position 0, and position 7 is never written—and its behaviour ranges widely between its two standard-order forms.

### Worked example 1: A simultaneous propagator by hand

Write an ECA rule in Wolfram’s standard neighbourhood order:

|               |     |     |     |     |     |     |     |     |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| position      | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
| neighbourhood | 111 | 110 | 101 | 100 | 011 | 010 | 001 | 000 |
| Rule 110      | 0   | 1   | 1   | 0   | 1   | 1   | 1   | 0   |

Thus Rule 110 is the byte 01101110. Let  $\sigma(p) = p+2 \pmod{8}$  on the displayed position indices: each source bit is complemented and written two positions further to the right, wrapping at the end. Carrying out the eight writes gives

$$P_{+2}(01101110) = 01100100,$$

which is Rule 100. This changes what the cell does to its black-and-white phenotype. For example, neighbourhood 011 maps to 1 under Rule 110 but to 0 under Rule 100. The propagator has not directly flipped a phenotype cell; it has changed the truth table that the cell will use.

### Definition 2: Genotype and phenotype

Each cell carries an eight-bit rule. The field of rules—the *genotype*—governs a field of black-and-white states—the *phenotype*—and neighbouring rules also interact. One complete local update (Box 2) evolves both layers; this is the computation behind the space–time diagrams in Figure 2.

### Worked example 2: One site, one complete MetaCA step

At time  $t$ , site  $x$  holds an eight-bit rule  $g_x(t)$  and a phenotype bit  $X_x(t)$ . One ordinary coupled step has three parts.

1. **Express the current rule.** The phenotype neighbourhood is evaluated by the rule at its centre. With numerical neighbourhood value  $n_x(t) = 4X_{x-1}(t) + 2X_x(t) + X_{x+1}(t)$ ,

its displayed position is  $7 - n_x(t)$ :

$$X_x(t+1) = g_x(t)(7 - n_x(t)).$$

2. **Blend neighbouring rules.** For each truth-table position  $j$ , inspect the three genotype bits  $(g_{x-1}(j), g_x(j), g_{x+1}(j))$ . If the two outer bits agree, copy their value. Thus  $(0, 1, 0)$  blends to 0. If they disagree, the centre rule decides by treating the triple as one of its ECA neighbourhoods, converting its numerical value  $n$  to displayed position  $7 - n$  as above. The eight coordinate-wise decisions form an intermediate rule

$$h_x = B(g_{x-1}(t), g_x(t), g_{x+1}(t)).$$

3. **Write one propagated bit.** Draw a source position  $K_x(t) \in \{0, \dots, 7\}$  uniformly and apply its elementary write to the intermediate byte:

$$g_x(t+1) = U_{\sigma, K_x(t)}(h_x).$$

The ordinary spatial runs make one such elementary write per cell and generation. The historical Figure 8 variant made two. By contrast,  $P_\sigma$  denotes the simultaneous eight-write map classified in *The Classification*; its fixed bytes are precisely the bytes left unchanged by every possible elementary write.

### Theorem 1

$P_\sigma$  has a fixed point (a fixed rule) if and only if every cycle of  $\sigma$  has even length. Since a one-cycle has odd length, this already requires  $\sigma$  to be fixed-point-free.

### Worked example 3: Even and odd cycles

For offset +2 on eight truth-table positions,

$$\sigma = (0\ 2\ 4\ 6)(1\ 3\ 5\ 7).$$

A fixed rule must change colour at every arrow because  $g(\sigma k) = \neg g(k)$ . Around either four-cycle it can therefore read 0101 or 1010: after four negations it returns consistently to its

starting bit. The two four-cycles are coloured independently, so two binary choices fix all eight truth-table entries  $\text{bit}[0], \dots, \text{bit}[7]$ . Read back as an ECA rule number—those eight bits in Wolfram neighbourhood order (the table of *The Object*) taken as an integer in 0–255—the four colourings are the fixed rules

$$2^2 = 4 : \quad 51, 102, 153, 204,$$

each with four zero-bits and four one-bits. An odd cycle admits no such colouring: after an odd number of negations the bit returns to its starting position demanding its own opposite, so no fixed rule exists. Admitting a fixed rule and settling onto one are distinct. An operator carrying an odd cycle has no fixed rule for its field to collapse onto, and its field correspondingly stays diverse (the any-odd reduced rotate+2, Figure 5, right); whether a fixed-point-bearing operator collapses onto one of its fixed rules is a separate, dynamical question (*Fixed Points Are Not Attractors*).

### Corollary 1

An all-even  $\sigma$  with  $k$  cycles has exactly  $2^k$  fixed bytes; any  $\sigma$  with an odd cycle has none.

### Definition 3: Langton's activity $\lambda$

$\lambda$  measures a rule's *activity*: the fraction of its neighbourhood entries that map to the non-quiescent state (Langton, 1990)—for an eight-bit ECA table, the Hamming weight over eight,  $\lambda = \|r\|_1/8$ , from 0 (every neighbourhood dies) through  $1/2$  to 1. A rule with  $\lambda = 1/2$  is *balanced*.

### Observation 2: Balance is a truth-table property

Langton's thesis was that complex behaviour concentrates near a critical  $\lambda$ , shown most clearly for larger tables ( $K = 4$ ,  $N = 5$ ); for the two-symbol, three-neighbour ECAs Langton is explicit that  $\lambda$  only roughly correlates with dynamics, and later work is cautious (Mitchell et al., 1993). Theorem 2 below concerns the *value*  $\lambda = 1/2$  itself—balance—which we keep separate from any empirical claim about what it predicts dy-

namically.

### Theorem 2

Every fixed point of every operator in the family has exactly four one-bits; equivalently, Langton's activity parameter  $\lambda = 4/8 = 1/2$ .

### Definition 4: Census protocol and aliveness

For each of the 40,320 permutations, we ran seeds 0–2 on a width-60 line for 100 generations. Initial rule bytes are independent uniform draws from 0–255, initial phenotype bits are independent fair binary draws, and both layers have fixed zero boundaries. Each genotype update uses the neighbour blend and one uniformly sampled elementary write described in Box 2. A layer's *change rate* is the mean fraction of cells differing between consecutive rows over the final 40 generations, averaged over the three seeds. The layer is *alive* when this rate is at least 0.10 and *dead* otherwise. Scoring genotype and phenotype separately gives four regimes: live/live, live/dead, dead/live, and dead/dead.

**Worked example 4: Reading the rotation survey**

The fields of Supplement 3, Figure 1 evolve under the *spatially coupled* dynamics of *Fixed Points Are Not Attractors*—stochastic elementary writes together with the coupling between cells—not  $P_\sigma$  applied cell-by-cell. This is what the survey turns on. Without the blend, Figure 5 already shows rotate+1 remaining active over the observed horizon while rotate+2 settles. With neighbour blending, the finite-horizon split persists and widens: the single 8-cycles— $\gcd(\text{offset}, 8) = 1$ , offset  $\pm 1$  and  $\pm 3$ —sustain a high-diversity field, while offset  $\pm 2$  ( $\gcd 2$ ) and  $\pm 4$  ( $\gcd 4$ ) collapse; that split is what the survey shows.

The two panels are *almost* mirror images, and where they differ is instructive. Offset  $+d$  and offset  $-d$  are the same permutation exactly when  $2d \equiv 0 \pmod{8}$ —so offset  $\pm 4$  (and the identity) are one operator with bit-identical panels, whereas offset  $\pm 1$  and  $\pm 3$  are distinct *inverse* rotations. Inverses share cycle structure, fixed rules, and the sustain-or-collapse verdict, but not their transient dynamics: offset  $+1$  and offset  $-1$  realise the same sustaining regime through visibly different fields.

**Definition 5: Temporal braid**

A *temporal braid* alternates two writings between complete coupled spatial updates: writing  $w_A$  is used at even times and  $w_B$  at odd times. More generally, a periodic *schedule* selects one writing at each time step. This preserves the intermediate spatial states and neighbour blends; it is not the algebraic composition  $P_{\sigma_A} \circ P_{\sigma_B}$  of two simultaneous maps.

**Definition 6: Interrupter**

An *interrupter* is an update between elementary propagator writes that is not itself a propagator write. The coordinate-wise neighbour blend of Box 2 is the interrupter in the ordinary spatial runs; a hold gate that suppresses a scheduled write is another example. An interrupter changes the coupled dynamics but does not, by definition, guarantee sustained diversity.

**Proposition 1: Composition parity**

Products alternate between complemented and uncomplemented coordinate permutations. Each operator of Equation (1) negates every bit ( $P_\sigma(g)[j] = \neg g[\sigma^{-1}(j)]$ , a coordinate permutation followed by a global flip), so under composition the flips cancel in pairs:  $P_\sigma \circ P_\tau$  is a *pure* coordinate permutation with no negation left—a bijection that fixes  $2^c$  bytes for  $c$  cycles but carries none of the forced  $\lambda = 1/2$  structure. An even number of propagators therefore composes to an uncomplemented coordinate permutation, and an odd number to another member of the permutation-propagator family.

**Definition 7: River**

A *river* is the composed operator obtained by following the collapsing offset-+2 propagator ( $\gcd 2$ ) with a phenotype-reading construction step—match the local phenotype against four template candidates, first match wins, and fall back to the all-zero rule when none matches.